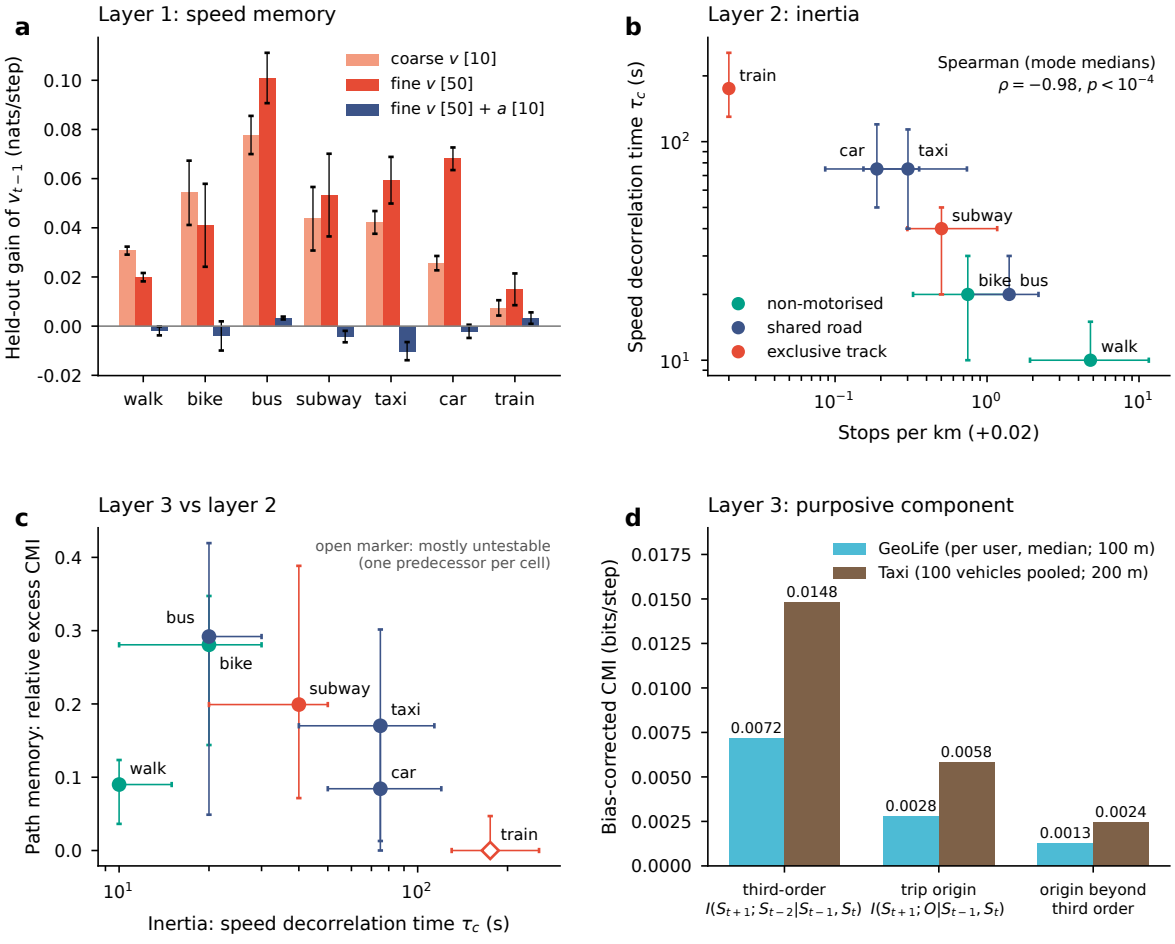
a
Layer 1: speed memory
coarse v [10]
fine v [50]
fine v [50] + a [10]
Held-out gain of v_{t-1} (nats/step)
0.10
0.08
0.06
0.04
0.02
0.00
-0.02
walk
bike
bus
subway
taxi
car
train
b
Layer 2: inertia
Spearman (mode medians)
$\rho = -0.98$, $p < 10^{-4}$
Speed decorrelation time τ_c (s)
10^2
10^1
train
car
taxi
subway
bike
bus
walk
non-motorised
shared road
exclusive track
10^{-1}
10^0
10^1
Stops per km (+0.02)
c
Layer 3 vs layer 2
open marker: mostly untestable
(one predecessor per cell)
Path memory: relative excess CMI
0.4
0.3
0.2
0.1
0.0
bus
bike
subway
taxi
walk
car
train
10^1
10^2
Inertia: speed decorrelation time τ_c (s)
d
Layer 3: purposive component
GeoLife (per user, median; 100 m)
Taxi (100 vehicles pooled; 200 m)
Bias-corrected CMI (bits/step)
0.0175
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
0.0148
0.0072
0.0058
0.0028
0.0024
0.0013
third-order
$I(S_{t+1}; S_{t-2}|S_{t-1}, S_t)$
trip origin
$I(S_{t+1}; O|S_{t-1}, S_t)$
origin beyond
third order